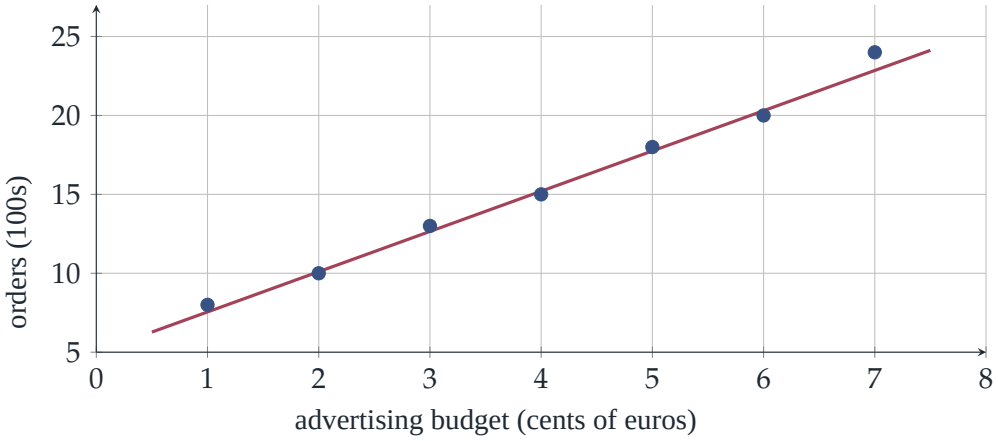
Chapter 5

Statistics with two quantitative variables

Competences: represent, choose an adjustment, interpolate, extrapolate with caution, communicate

5.1 Reference course

A two-variate statistical series consists of couples (x_i, y_i). The cloud of points allows to observe a trend, dispersion and possible atypical values.



Key ideas

A straight line of adjustment $y = ax + b$ provides estimates, not exact equities.
a described the estimated mean change in y when x increases by one unit.
remains in the range of the observed values; extrapolation emerges from this interval and must be presented with caution.

The average point $G(\bar{x}, \bar{y})$ a for coordinates $\bar{x} = \frac{x_1 + \cdots + x_n}{n}$ and $\bar{y} = \frac{y_1 + \cdots + y_n}{n}$. A right of regression obtained at the calculator passes through G, to the nearest round.

Method

To exploit $y = ax + b$: 1) replace x by the value requested to estimate y ; 2) resolve $ax + b = y_0$ to estimate x ; 3) round according to context; 4) specify interpolation or extrapolation ; 5) to remind that the result depends on the model.

A change in variable, for example $z = \log(y)$ or $z = \frac{1}{y}$, can make a cloud approximate- We'll adjust it then. (x, z), then we go back to the variable y.

Meeting 31 — Dot cloud

Two-variate statistics

26/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Build a cloud.

· Choose scales.

· Identify variables.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase . . . . . .

2. Resolve 3 $x + 4 = 25$. . . . . .

3. Calculate the average of 6, 8 and 10 . . . . . .

4. Give the slope between $(1, 3)$ and $(4, 9)$. . . . . .

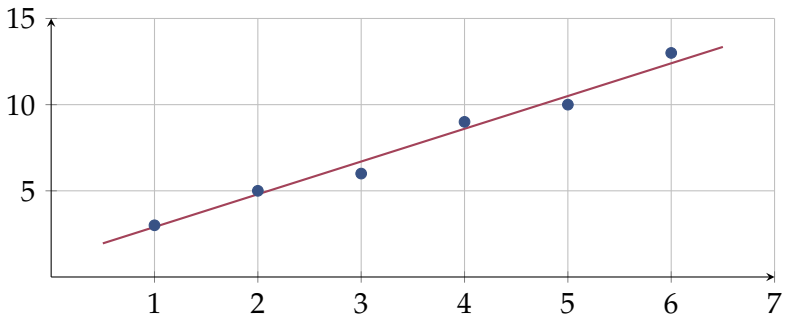
5. With $y = 2x + 5$, calculate y for $x = 7$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: build a cloud and choose the scales.

Research activity

The table (x_i, y_i) gives the Expenditure advertising and the sales of Six months : $(1, 12), (2, 15), (3, 17), (4, 22), (5, 24), (6, 28)$. Choosing scales, plotting the cloud and placing the average point.



Trace written

A cloud of dots represents couples (x_i, y_i). The axes, units and scales shall be indicated ; each couple gives only one point.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. For $y = 3x + 13$, estimate y where $x = 8$.
2. Resolve 3 $x + 13 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, courses p. 110·113; exercises p. 114, n° 1 to 6.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 114, exercise 2

Represent the cloud of points in the next statistical series in a plan coordinate system.

x_i	-3	0	3	5	8
y_i	1	-1	-2	5	7

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 32 — Reading and interpreting the cloud

Two-variate statistics

28/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Describe a trend.

· Identify dispersal and atypical.

· Avoid abusive causality.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 7 - x + 2$. .....
2. Calculate the average of 7, 9 and 11 .....
3. Give the slope between $(1, 3)$ and $(4, 9)$. .....
4. With $y = 2x + 5$, calculate y for $x = 7$. .....
5. Resolve 3 $x + 4 = 25$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: describe a trend and identify dispersion and Atypical.

Research activity

Compare three clouds: tight increasing trend, dispersed decreasing trend, absence of
Write two sentences for each without using the word "cause".

Trace written

Describe a cloud consists of commenting on direction, form, dispersion and atypical values, without asserting an unestablished causality.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. For $y = 4x + 14$, estimate y where $x = 8$.
2. Resolve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 116, Nos. 11 to 15. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the

digital resources of the manual.

Manuel Calao · p. 114, exercise 6

On the chart of the manual is shown the number of cigarettes sold in France according to the number of wolf packs recorded in France between 1995 and 2017.
How can we describe the evolution of cigarette sales from this graph?
So much, can we think that there is a connection between these facts?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 33 — Linear fit

Two-variate statistics

02/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Get an affine adjustment.

· Interpret a and b.

· Check with the average point.

Fluency · 6 to 8 minutes, without calculator

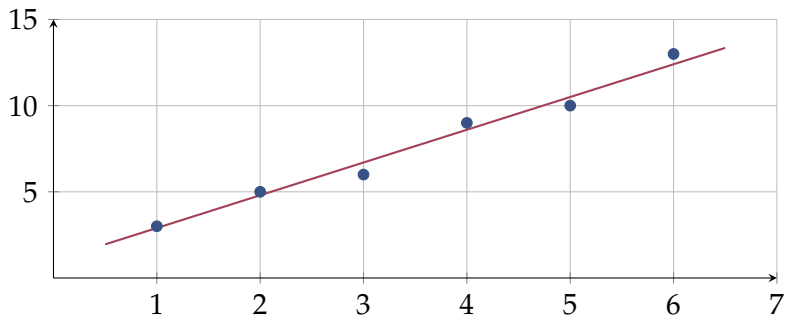
1. Calculate 20 per cent of 160 .....
2. Give the slope between $(1,3)$ and $(4,9)$. .....
3. With $y = 2x + 5$, calculate y for $x = 7$. .....
4. Resolve 3 $x + 4 = 25$. .....
5. Calculate the average of 8, 10 and 12 .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: get a fine tuning and interpret a and b.

Research activity

With data from session 31, get regression refines to the calculator. Interpret slope and verify that the average point belongs approximately to the right.



Trace written

A Right $y = ax + b$ Adjust a cloud. a represents the estimated average change in y for one additional unit of x.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. For $y = 5x + 15$, estimate y where $x = 8$.
2. Resolve 5 $x + 15 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, courses p. 110·113; exercises p. 117, n° 16 to 18.
session is identified below by its page and number; the reference also allows you to find
the figures and digital resources of the manual.

Manuel Calao · p. 115, exercise 9

1. Represent in a plan coordinate system the point cloud associated with the statistical series opposite.

x_i	-2	-1	0	2	3	5
y_i	4	8	12	16	16	20

Draw the right of equation $y = 2x + 10$ in this coordinate system. The adjustment of the cloud by this right
Is it satisfactory? If not, propose another one whose equation will be specified
reduced.

2. Using this equation, determine:

· the order of the point of the right of adjustment having for abscisse 1 ;

· the order of the point of the right of adjustment having for abscisse 10;

· abscisse from the point of the right of adjustment having as an order 7.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and
indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 34 — Interpolation

Two-variate statistics

04/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Interpolate.

· Solve with the model.

· Express an estimate.

Fluency · 6 to 8 minutes, without calculator

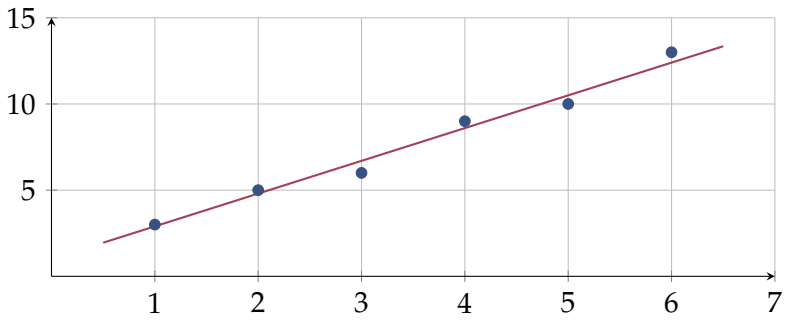
1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. With $y = 2x + 5$, calculate y for $x = 7$. . . . . .
3. Resolve 3 $x + 4 = 25$. . . . . .
4. Calculate the average of 9, 11 and 13 . . . . . .
5. Give the slope between $(1, 3)$ and $(4, 9)$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: interpolate and resolve with the model.

Research activity

Over the period observed $x \in [1, 12]$ the model is $y = 2{,}8x + 14$. Estimate y for $x = 7{,}5$, then estimate x where $y = 35$. Qualify each estimate.



Trace written

An interpolation estimates a value within the observed interval.
model-dependent estimate.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. For $y = 6x + 16$, estimate y where $x = 8$.
2. Resolve 6 $x + 16 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 118, n° 19 to 24. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 115, exercise 8

The table below shows the number of new electric vehicles registered in France.
2014 à 2018.

Year		2014	2015	2016	2017	2018
Rank of year	x_i	0	1	2	3	4
Thousands of vehicles	y_i	10,5	17,3	21,7	24,9	31

1. Represent in a coordinate system the point cloud associated with this time series.
2. Propose a fine tuning of this cloud.
3. If it is assumed that this adjustment model remains valid for several years, how many vehicles would have been registered in 2020? From which year would it have exceeded 50,000 New vehicles registered?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 35 — Extrapolation

Two-variate statistics

09/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Extrapolate carefully.

· Evaluate plausibility.

· Present a model boundary.

Fluency · 6 to 8 minutes, without calculator

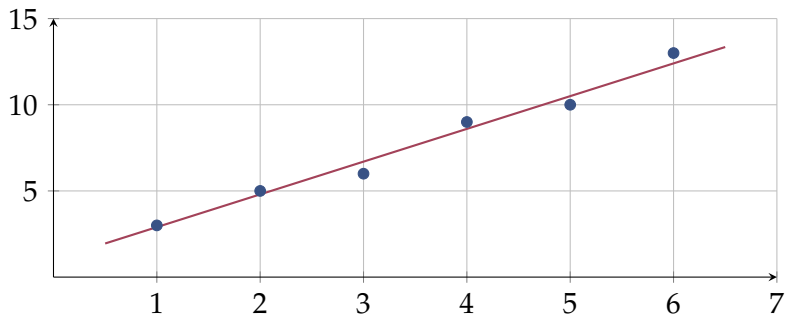
1. Reduce 3 $x + 3 - x + 2$. .....
2. Resolve 3 $x + 4 = 25$. .....
3. Calculate the average of 3, 5 and 7 .....
4. Give the slope between $(1, 3)$ and $(4, 9)$. .....
5. With $y = 2x + 5$, calculate y for $x = 7$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: carefully extrapolate and evaluate plausibility.

Research activity

The same model $y = 2{,}8x + 14$, adjusted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it can be unreliable.



Trace written

Extrapolation is estimated to be outside the observed range.
necessary.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. For $y = 2x + 12$, estimate y where $x = 8$.
2. Resolve 2 $x + 12 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 119, Nos. 25 to 27. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 118, exercise 20

The annual turnover of an enterprise (in EUR '000) was increased during its five years of operation. the first years of existence.

Year		2016	2017	2018	2019	2020
Rank of year	x_i	0	1	2	3	4
Turnover	y_i	315,6	489,9	579,3	763,5	874,5

1. Representing, in an orthogonal coordinate system, the cloud of points $M_i(x_i; y_i)$ of this statistical series.
2. We adjust this cloud of points from the right (M_1M_5). Determine an equation of this line (round the coefficients to the tenth).
3. If the trend continues, what turnover can be expected for 2024? What year will the turnover exceed ·2 million?

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 36 — Change of variable

Two-variate statistics

11/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Make a variable change.

· Adjust new data.

· Return to the initial variable.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 80 .....

2. Calculate the average of 4, 6 and 8 .....

3. Give the slope between $(1,3)$ and $(4,9)$. .....

4. With $y = 2x + 5$, calculate y for $x = 7$. .....

5. Resolve 3 $x + 4 = 25$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: make a change of variable and adjust the New data.

Research activity

The data seems exponential. $z = \log(y)$ and adjustment gives $z = 0{,}08x + 1{,}2$.
Express y according to the x, then calculate y for $x = 5$.

Trace written

A variable change can align a cloud. After adjustment in the new variable, it must go back to the initial variable.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. For $y = 3x + 13$, estimate y where $x = 8$.

2. Resolve 3 $x + 13 = 50$.

3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 120, n° 28·29. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the

digital resources of the manual.

Manuel Calao · p. 120, exercise 29

In a concert hall, a reading of sound intensity levels from a power generator
The measurements were recorded in the table below.
below.

Distance x_i (m)	0,5	1	2	3	5	10	11	12	15	20
Sound intensity y_i (dB)	97	85	73	66	59	53	55	52	55	49

An affine adjustment does not seem relevant, a change in variables is envisaged.

1. Complete the table of the manual with $u_i = \log(x_i)$.
2. Represent the dot cloud $(u_i; y_i)$ in a coordinate system, then propose an affine adjustment.
3. Deduce an expression of sound intensity as a function of distance.
4. From the expression found, determine the expected sound intensity at a distance of 8 meters, then 30 meters.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 37 — Model selection and adjustment

Two-variate statistics

02/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Compare models.

· Use residues.

· Define a field of use.

Fluency · 6 to 8 minutes, without calculator

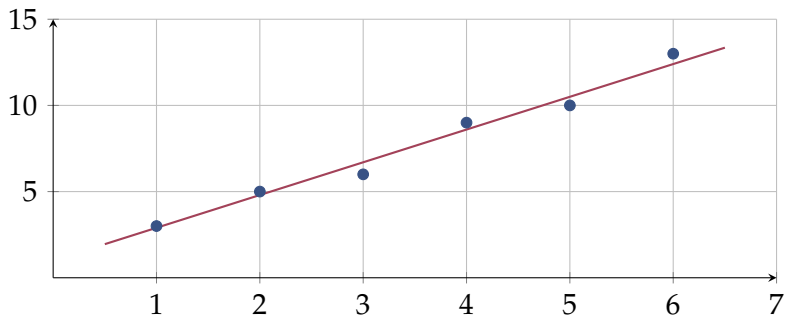
1. Write the multiplier coefficient of a 10 per cent increase . . . . . .
2. Give the slope between $(1,3)$ and $(4,9)$. . . . . .
3. With $y = 2x + 5$, calculate y for $x = 7$. . . . . .
4. Resolve 3 $x + 4 = 25$. . . . . .
5. Calculate the average of 5, 7 and 9 . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: compare models and use residues.

Research activity

Two models give total absolute residues 18 ,4 and 11,7 on the observed area.
However, there is an unrealistic growth after $x = 20$. Choosing a model to interpolate $x = 12$, then discuss extrapolation to $x = 50$.



Trace written

A model is chosen according to the shape of the cloud, the size of the residues and the field of use.
best visual fit does not allow unlimited extrapolation.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. For $y = 4x + 14$, estimate y where $x = 8$.
2. Resolve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 121, No. 30 to 32, and p. 122, No. 33.
is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 119, exercise 25

The determination of the caffeine concentration of a drink by measure of absorbance is sought. This, we use a laser beam and we measure the attenuation of the light emitted after crossing the liquid. For the purposes of the experiment, several solutions of which the caffeine content is known are tested and the following set of values is obtained.

Absorbance A	0,155	0,320	0,465	0,630	0,935
Concentration C (mg/L)	5	10	15	20	30

1. Represent this data using a cloud of points in a plan coordinate system.
2. Does an adjustment seem relevant here? If applicable, express the value C of the caffeine concentration as a function of value A Absorbance.
3. For a beverage containing caffeine an absorbance of 0 is measured. ,5. What then is its Caffeine concentration?

Personal work

Resuming Exercise 2 with the following data:
305 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 38

Assessment and evaluation

Two-variate statistics

04/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Choose and operate an adjustment.

· Write a statistical conclusion.

· Analyze an error.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 6 - x + 2$. .....
2. With $y = 2x + 5$, calculate y for $x = 7$. .....
3. Resolve 3 $x + 4 = 25$. .....
4. Calculate the average of 6, 8 and 10 .....
5. Give the slope between $(1, 3)$ and $(4, 9)$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: choose and exploit an adjustment and write a Statistical conclusion.

Research activity

From a cloud provided by the teacher, produce a complete answer: choice of model, Equation, estimate, type of estimate, unit and reserve. Exchange copies and use grid of six criteria.

Trace written

A complete statistical response contains the model, calculation, rounding, unit, type of estimate and a reservation on validity.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. For $y = 5x + 15$, estimate y where $x = 8$.
2. Resolve 5 $x + 15 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Reference in the manual

Supplements: Calao, p. 122·123. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 118·119, exercise 24

Choose the right proposal. Justify the answer. We consider the cloud of points $M_1(1;16)$, $M_2(3;13)$ and $M_3(5;4)$.

1. The average point of this cloud is:
   a. the point of contact $(3;10)$;
   b. the point M_2 ;
   c. the point of contact $(3;11)$.
2. The Right (M_1M_3) a for equation:
   a. $y = 4x + 16$;
   b. $y = -3x + 19$;
   c. $y = -2{,}93x + 17{,}7$.
3. The right of adjustment obtained by the calculator:
   a. go through one of the three points;
   b. is parallel to the right (M_1M_3) ;
   c. goes through the origin of the coordinate system.

Personal work

Resuming Exercise 3 with the following data:
310 and rate or common difference / ratio by 4. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.